LAPORAN DEVELOPER DAN OPERASIONAL

MINGGU KE - 12

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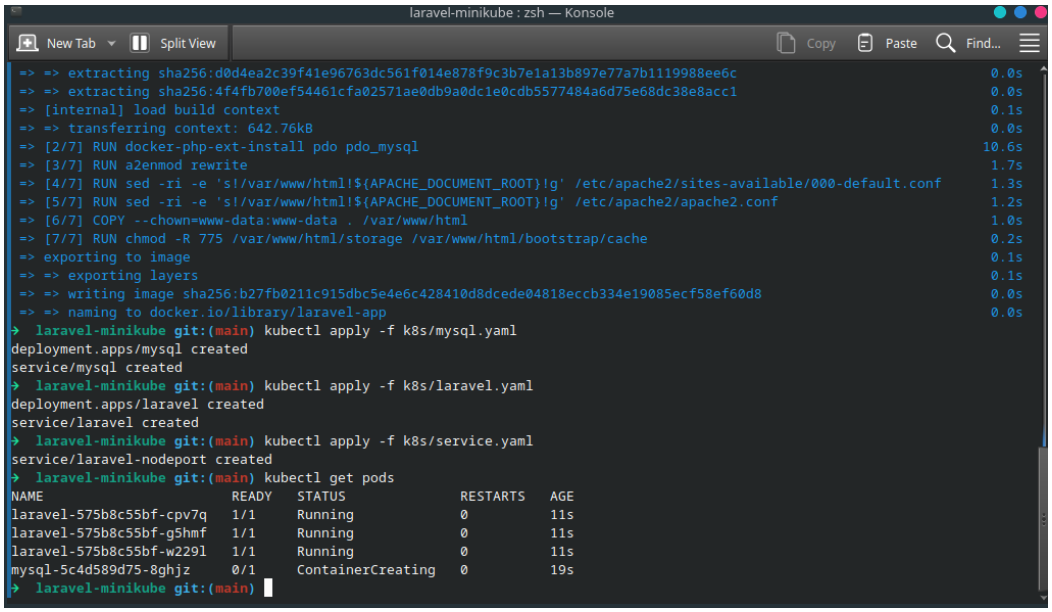
1. Konfigurasi Dockerfile

Docker file berisi kode untuk instalasi image PHP 8.3 dengan Apache, lalu menginstal ekstensi MySQL, mengaktifkan modul rewrite URL, dan mengarahkan root dokumen ke folder public

Perintah kubectl get nodes digunakan untuk memastikan cluster sudah siap. Jika muncul status "Ready", berarti minikube kita sudah bisa digunakan.



2. Build dan Load Image ke Minikube

Dari Dockerfile yang sudah kita buat, kita membuat image bernama laravel-app menggunakan perintah *docker build*. Image ini masih ada di Docker Desktop, belum masuk ke Minikube, kita perlu memasukkan image ke minikube dengan perintah *image load*

```
laravel-minikube : zsh — Konsole
New Tab   Split View                                   Copy   Paste   Find...
➜  laravel-minikube git:(main) eval $(minikube docker-env)
➜  laravel-minikube git:(main) docker build -t laravel-app .
[+] Building 69.4s (13/13) FINISHED                                                    docker:default
 => [internal] load build definition from dockerfile                                              0.0s
 => => transferring dockerfile: 546B                                                              0.0s
 => [internal] load metadata for docker.io/library/php:8.3-apache                                 4.0s
 => [auth] library/php:pull token for registry-1.docker.io                                        0.0s
 => [internal] load .dockerignore                                                                 0.0s
 => => transferring context: 2B                                                                   0.0s
 => [1/7] FROM docker.io/library/php:8.3-apache@sha256:e1c7243a9fa8ab2996e1972800baa5d4874a5a1e67a9df34f2aa57482fb27d  48.9s
 => => resolve docker.io/library/php:8.3-apache@sha256:e1c7243a9fa8ab2996e1972800baa5d4874a5a1e67a9df34f2aa57482fb27d6  0.0s
 => => sha256:dff6526ccc462dde7d4ca33381c692b78022bc246df8458eebcb72bb8654e5ff 3.82kB / 3.82kB    0.0s
 => => sha256:13115c44ac8e17e29d300feddf8d25ffabf840916d8a28249a7770f89aaf3a50 11.84kB / 11.84kB  0.0s
 => => sha256:a923b3cdaa21f3c6e30c9662c5be29400c6b6f28e913ffb7c714816ce30bc886 226B / 226B        1.8s
 => => sha256:e1c7243a9fa8ab2996e1972800baa5d4874a5a1e67a9df34f2aa57482fb27d66 10.35kB / 10.35kB  0.0s
 => => sha256:f7417fc69aeb1abfb52a71b35ffe493f50ca3ff41e13bd8ce993e487500336ff 117.84MB / 117.84MB 45.3s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB 39.7s
 => => sha256:a5f587e05427862542219f5f49fab984a5a449fb100c7eff120cc3f372f72ac2 225B / 225B        3.4s
 => => sha256:899cbc2fdb85666778ad13381bbb12b2ade6a9c537fa794cb22d40a887722d0b 4.23MB / 4.23MB   14.0s
 => => sha256:cac4a54c06dfc039dccbf6901b77f09f00b5e05ae1e35f71dfb583b194f34156 430B / 430B       14.8s
 => => sha256:c32b51c9e3e908184961502dfbb02452fe42913ce5ab2837897bfac0927604e4 483B / 483B       15.7s
 => => sha256:27b0fec9d44b5a409c6da0cb2c4ba39287b5b7c04542cebfa7f37b53dcebaa31 12.77MB / 12.77MB 30.3s
 => => sha256:9f7cb9638f2e12c44aaf249c640506296dd0a6ca6b2e49bd2b9139e7f963a06d 486B / 486B       31.1s
 => => sha256:c9acb28d1f7020d8f94b61c24b781a837959cdc7fa878d4f15ef746df237f1c0 11.71MB / 11.71MB 36.3s
 => => sha256:bd5f94b940435188be209dde1f683295851ae5ce9ea4f5b5dd9ceb511e0ca894 2.46kB / 2.46kB   37.3s
 => => sha256:79b094a6a742b7fdc596f15854cb257b1311d40d230db3e29b41222b2dd333ba 248B / 248B       38.3s
 => => sha256:2a0b801d7f1e3d7b1d2dd45bf832c664bc23d3d66ff9d8d109f3b224ce9f78fd 243B / 243B       39.3s
 => => sha256:d0d4ea2c39f41e96763dc561f014e878f9c3b7e1a13b897e77a7b1119988ee6c 889B / 889B       40.2s
 => => extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd          0.6s
```

3. Menjalankan Aplikasi Laravel di Kubernetes

Sebelum menjalankan Laravel, kita perlu migrasi database dulu ke Laravel di Kubernetes dengan cara masuk ke dalam pod Laravel dengan perintah kubectl exec –it, setelah masuk kita lakukan php artisan migrate untuk migrasi database.

```
➜  laravel-minikube git:(main) kubectl exec -it laravel-775458d668-9ph5w -- php artisan migrate --force

  INFO  Preparing database.

  Creating migration table ..................................................... 7.63ms DONE

  INFO  Running migrations.

  0001_01_01_000000_create_users_table ........................................ 29.50ms DONE
  0001_01_01_000001_create_cache_table ........................................ 21.44ms DONE
  0001_01_01_000002_create_jobs_table ......................................... 24.96ms DONE

➜  laravel-minikube git:(main) minikube service laravel-nodeport
```

NAMESPACE	NAME	TARGET PORT	URL
default	laravel-nodeport	80	http://192.168.49.2:30007

```
🎉  Opening service default/laravel-nodeport in default browser...
```

4. **Laravel pada browser**

Secara otomatis setelah menjalankan perintah tersebut akan muncul